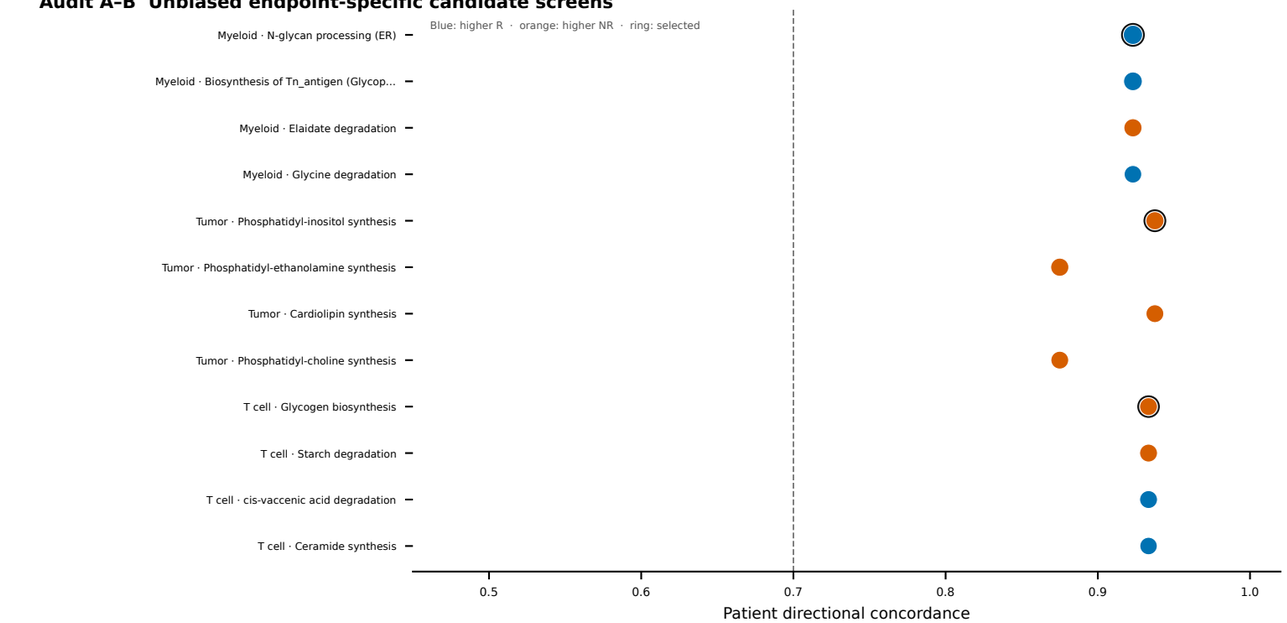
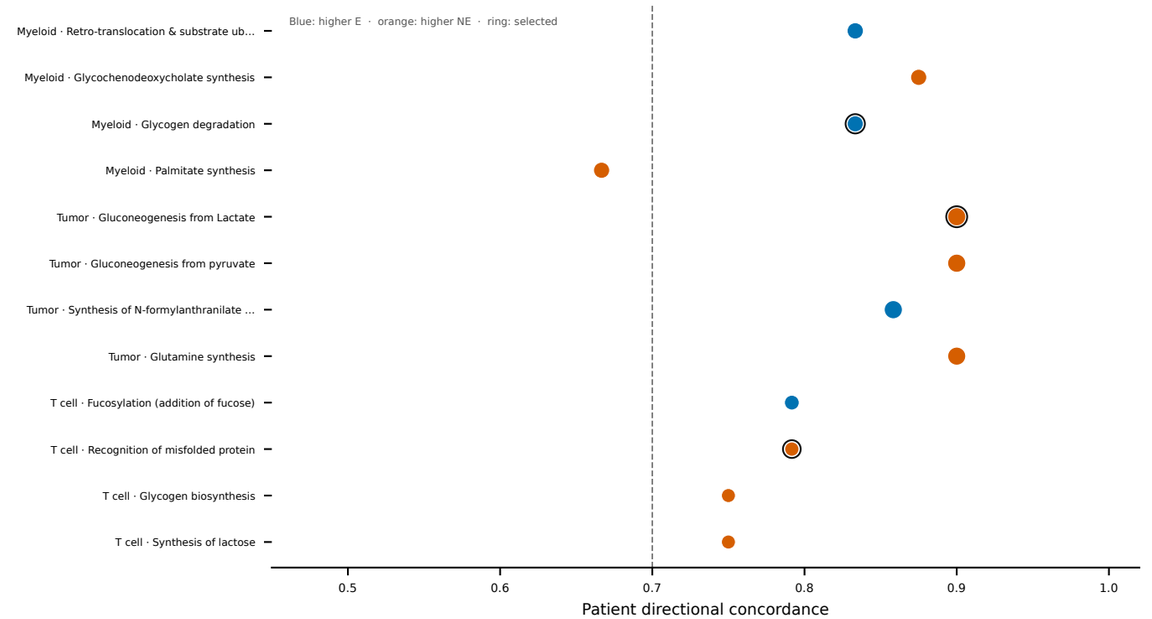
Figure S3_A Candidate selection and patient-level statistical audit

Audit A–B Unbiased endpoint-specific candidate screens

A Clinical R/NR candidates

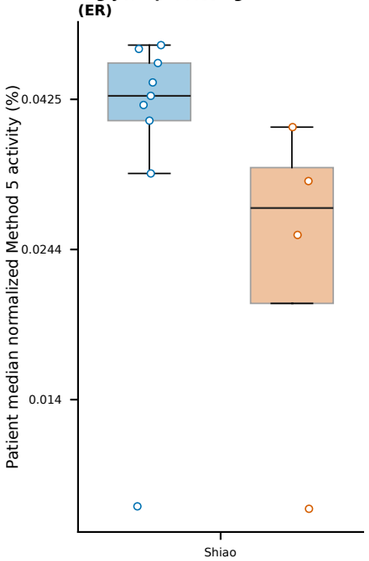


B Clonotype E/NE candidates



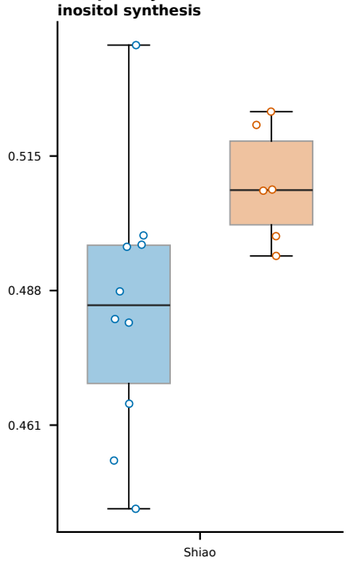
Audit C–H Patient-level selected-task distributions

C Myeloid · R/NR N-glycan processing (ER)



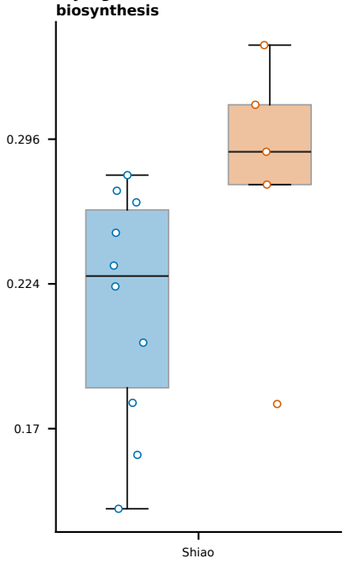
δ=-0.78; p=0.095; q=1

D Tumor/epithelial · R/NR Phosphatidyl-inositol synthesis



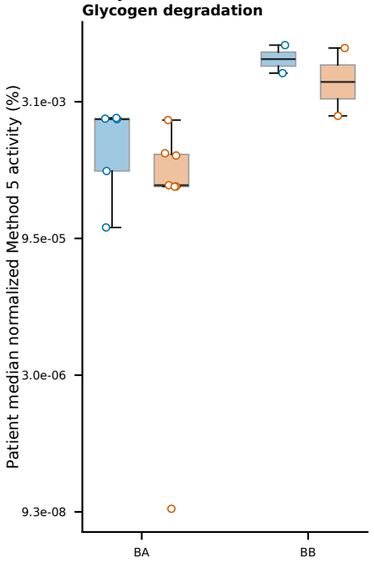
δ=+0.67; p=0.032; q=1

E T cell · R/NR Glycogen biosynthesis



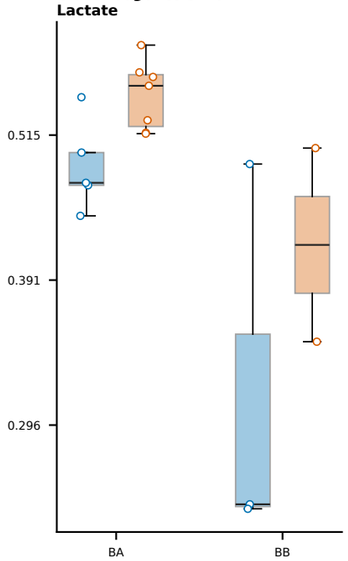
δ=+0.64; p=0.21; q=1

F Myeloid · E/NE Glycogen degradation



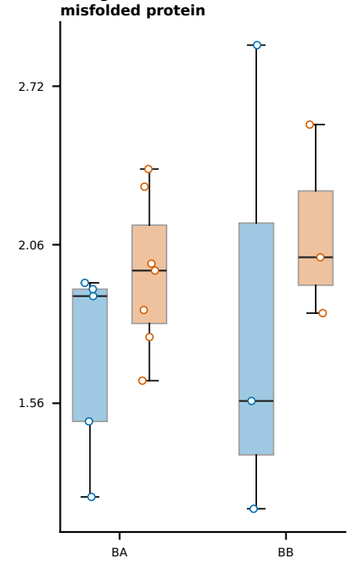
BA: δ=-0.49, q=0.67
BB: δ=-0.50, q=1

G Tumor/epithelial · E/NE Gluconeogenesis from Lactate



BA: δ=+0.83, q=0.59
BB: δ=+0.67, q=1

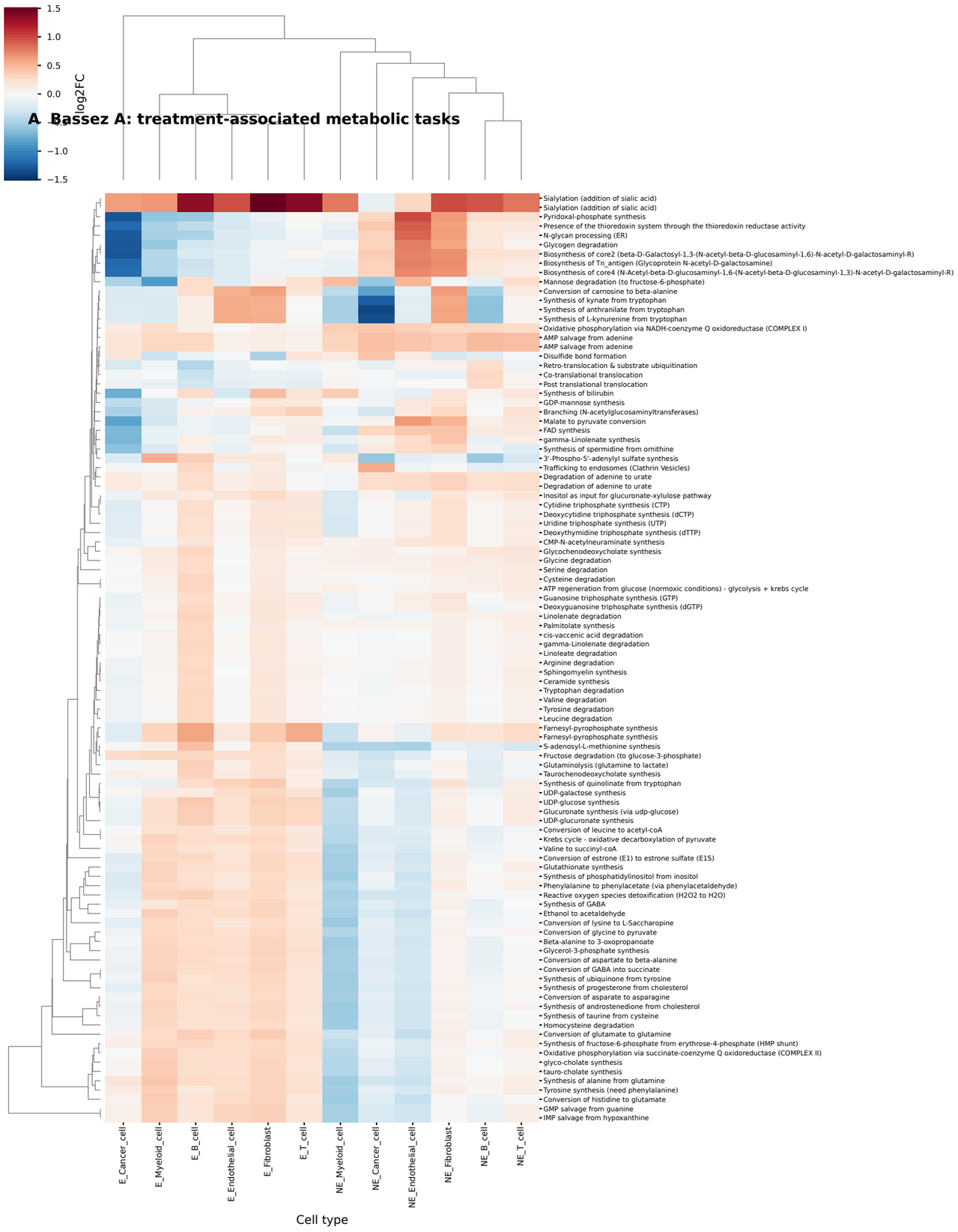
H T cell · E/NE Recognition of misfolded protein



BA: δ=+0.49, q=0.8
BB: δ=+0.33, q=1

Figure S3_B Cohort-paired treatment-associated metabolic-task and GO activity patterns

A Bassez A: treatment-associated metabolic tasks



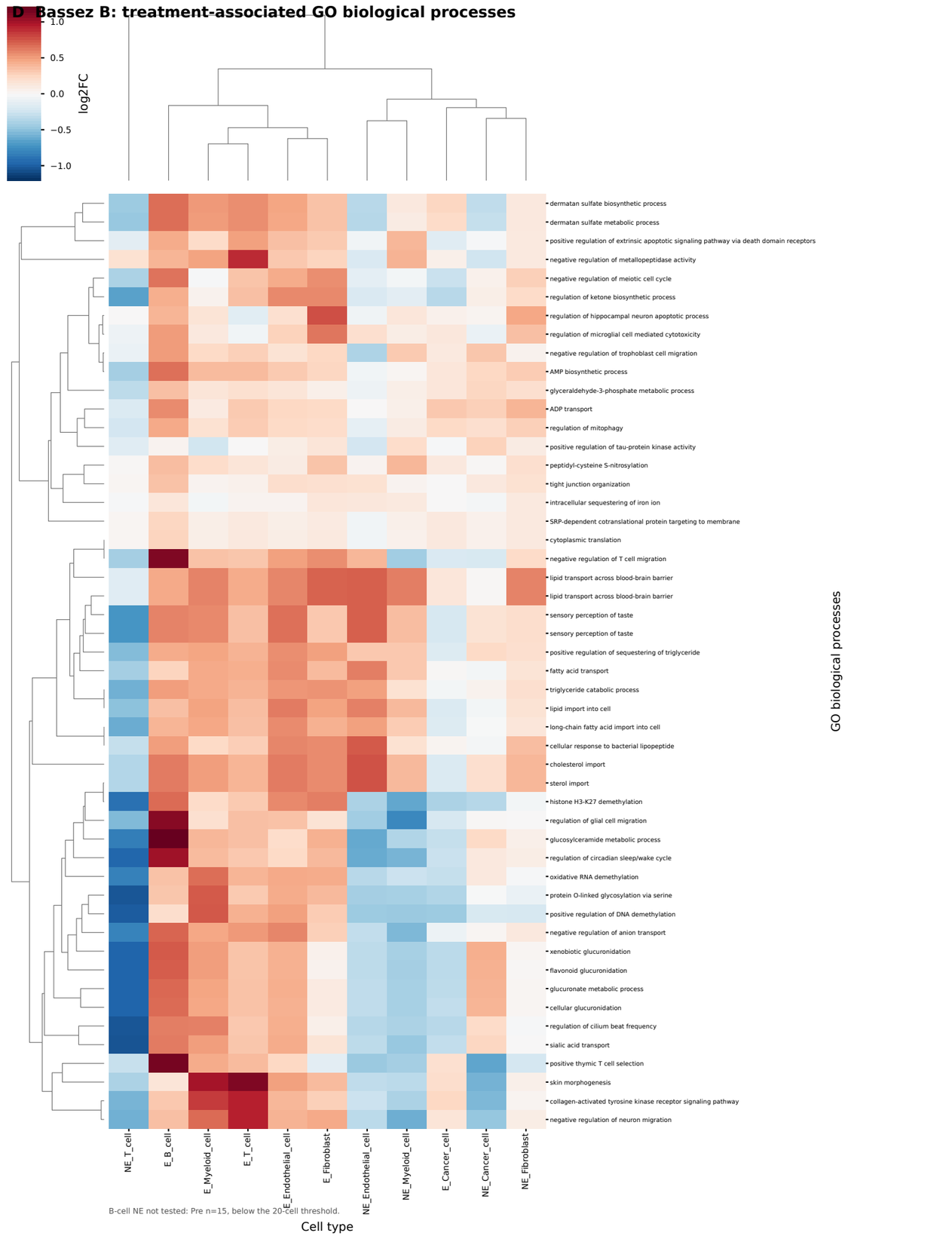
B Bassez A: treatment-associated GO biological processes



C Bassez B: treatment-associated metabolic tasks



D Bassez B: treatment-associated GO biological processes

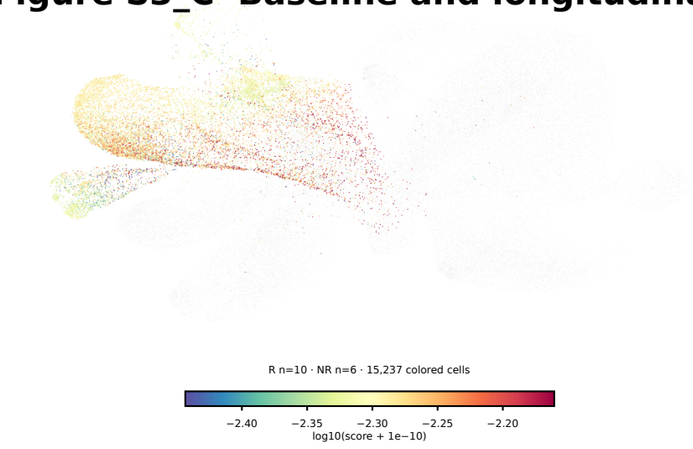


B-cell NE not tested: Pre n=15, below the 20-cell threshold.

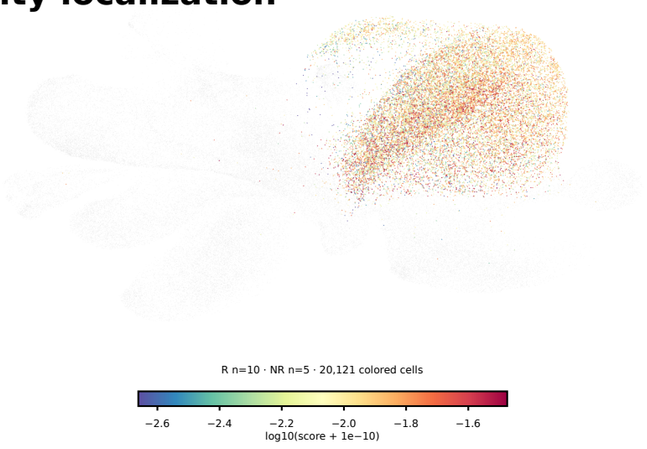
Figure S3_C Baseline and longitudinal task-activity localization

Baseline A–F

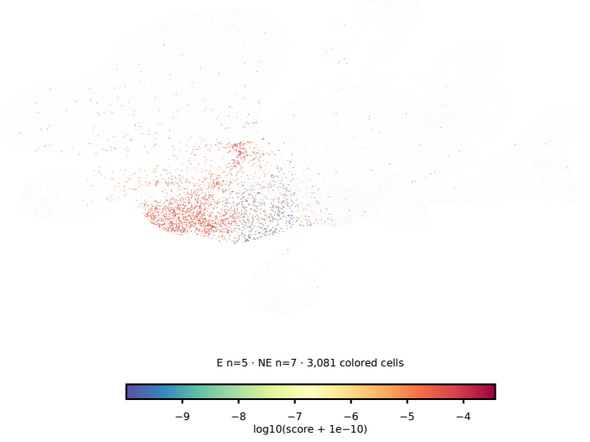
A E · Shiao · Tumor/epithelial
Base · Arachidonate synthesis



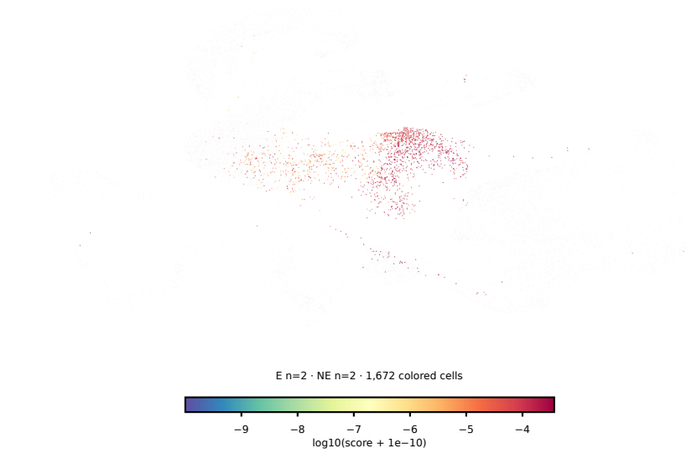
B F · Shiao · T cell
Base · ERAD / recognition of misfolded protein



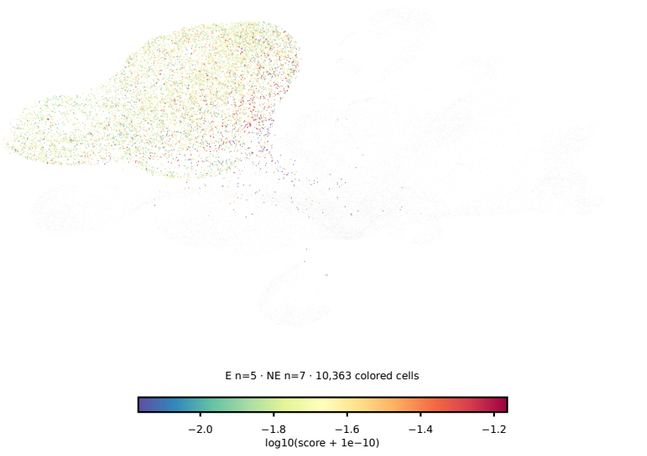
C G · Bassez A · Myeloid
Pre · Glycogen degradation



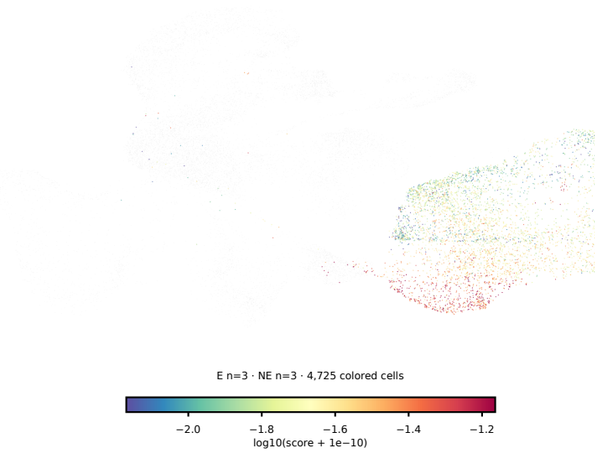
D G · Bassez B · Myeloid
Pre · Glycogen degradation



E H · Bassez A · T cell
Pre · ERAD / recognition of misfolded protein



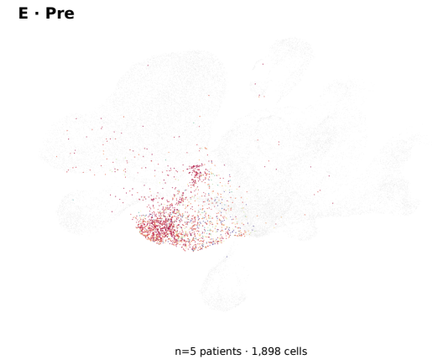
F H · Bassez B · T cell
Pre · ERAD / recognition of misfolded protein



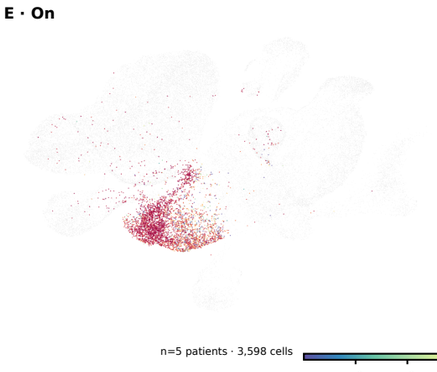
Longitudinal A–D

A I · Bassez A · Myeloid · Glutathione synthesis

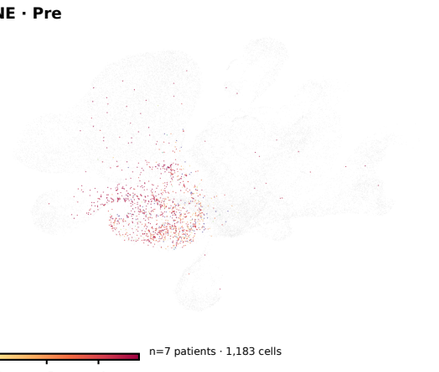
E · Pre



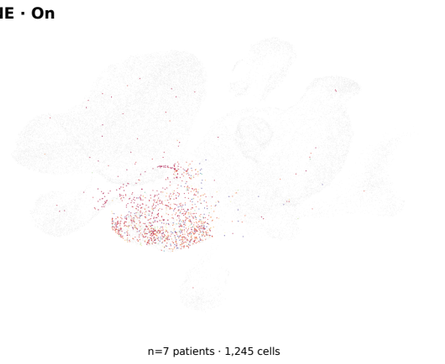
E · On



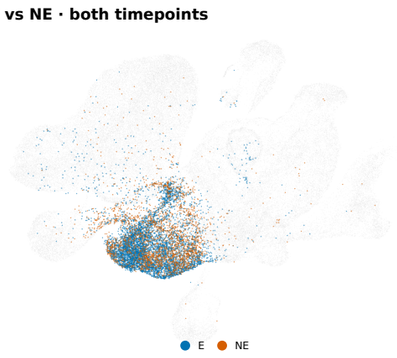
NE · Pre



NE · On

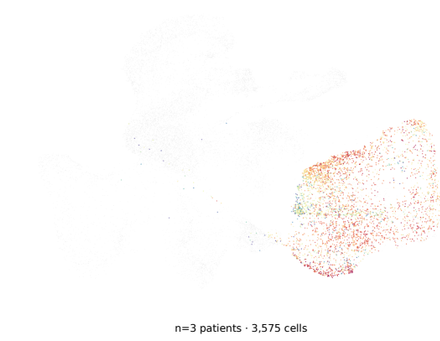


E vs NE · both timepoints

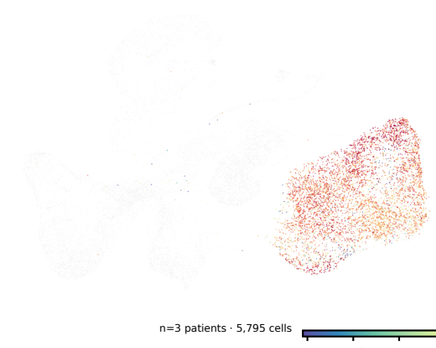


B J · Bassez B · T cell · Gluconeogenesis from glutamine

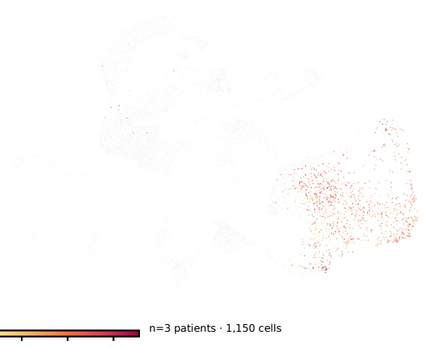
E · Pre



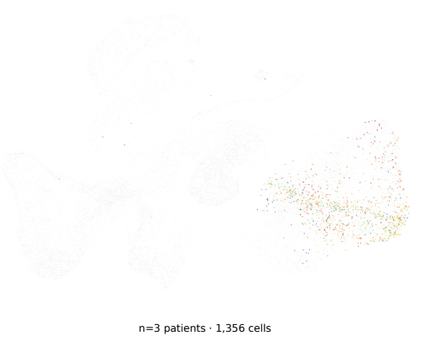
E · On



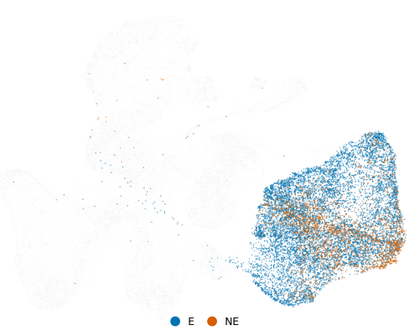
NE · Pre



NE · On



E vs NE · both timepoints



C K · Shiao · Fibroblast · Alanine synthesis from glutamine

R · Base



R · RTPD1



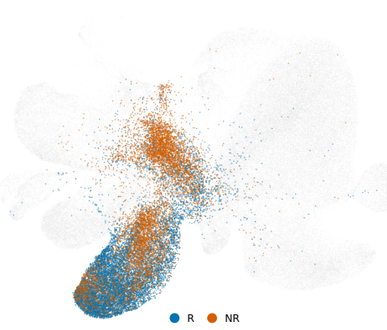
NR · Base



NR · RTPD1

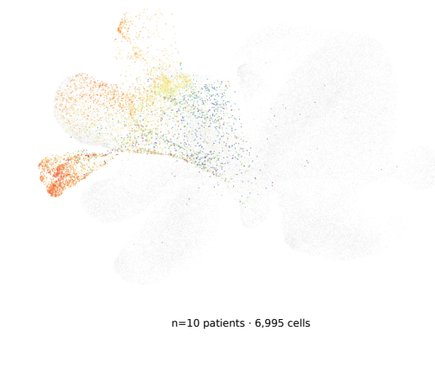


R vs NR · both timepoints

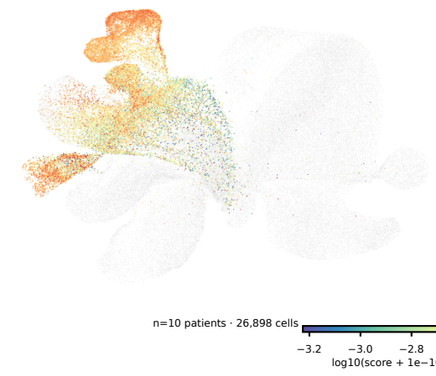


D L · Shiao · Tumor/epithelial · Bilirubin synthesis

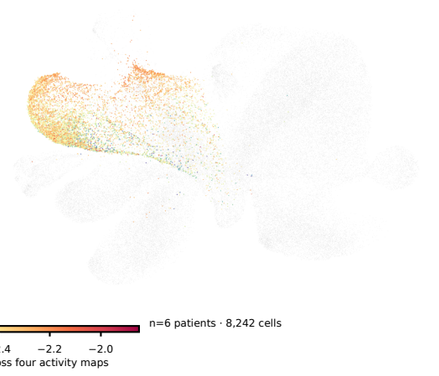
R · Base



R · PD1



NR · Base



NR · PD1



R vs NR · both timepoints

